Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

Table of Contents

1. Introduction 3
2. Problem Context and Objective 3
3. Dataset Description 3
4. Data Preparation and Feature Engineering 4
5. Exploratory Data Analysis 5
6. Clustering Methodology 8
7. Cluster Evaluation and Validation 11
8. Cluster Profiles and Insights 12
Cluster 0: Premium Large-Property Buyers 12
Cluster 1: Older Multi-Property Buyers 13
Cluster 2: Dissatisfied Mid-Value Buyers 13
Cluster 3: Satisfied Value-Oriented Buyers 13
Overall insight from the segmentation 13
9. Business Recommendations 14
Recommendation 1: Prioritize premium targeting for Cluster 0 14
Recommendation 2: Build retention and loyalty strategies for Cluster 1 14
Recommendation 3: Treat Cluster 2 as a service-improvement priority 15
Recommendation 4: Maintain and scale engagement for Cluster 3 15
Recommendation 5: Do not over-segment by client type or loan status alone 15
10. Limitations 15
11. Conclusion 16

1. Introduction

The real estate market involves highly diverse buyer groups with varying motivations, financial behavior, and regional preferences. Buyers may differ in terms of whether they purchase for home ownership or investment, whether they rely on financing, what types of properties they prefer, and how frequently they transact. In the absence of segmentation, real estate firms risk treating all buyers as one broad category, which can result in inefficient marketing, weak customer targeting, generic property recommendations, and missed investment opportunities. These business challenges are central to the Parcl project context.

This project applies machine learning-based buyer segmentation and investment profiling to support real estate market intelligence. The aim is to identify hidden patterns in buyer behavior and translate them into meaningful customer segments that can support improved targeting, better investment insight, and more data-driven decision-making. The study focuses on clustering clients based on demographic, behavioral, financing, and property-related variables, and then interpreting the resulting segments in business terms.

2. Problem Context and Objective

According to the project requirements, Parcl currently lacks a data-driven understanding of different types of property buyers, investment motivations across demographics, geographic differences in investment behavior, and customer financing patterns. This leads to inefficient marketing spend, generic property recommendations, and poor investor targeting.

The objective of this project was therefore to build a buyer segmentation framework using clustering techniques. More specifically, the project aimed to:

- identify naturally occurring customer groups in the buyer base
- profile these groups using client and property-level features
- validate the clustering structure using more than one clustering method
- generate business insights and recommendations for marketing and investment strategy

3. Dataset Description

The analytical workflow began with a client-level dataset containing fields such as client type, gender, country, region, date of birth, acquisition purpose, loan application status, referral channel, and satisfaction score. These variables reflect customer demographics, financing behavior, and acquisition context, and form the foundation of the buyer segmentation exercise.

To strengthen the segmentation, a second dataset containing property transaction information was integrated into the analysis. This included attributes such as transaction date, unit category, floor area, sale price, listing status, and client reference. Because property records existed at transaction level rather than client level, the property dataset was aggregated into client-level summary measures before merging. These engineered features included the number of sold properties linked to each client, average sale price, total sale price, average floor area, most recent transaction date, and counts of apartment and office purchases.

A derived age variable was also created from the client date of birth field to make age directly usable in analysis and clustering.

4. Data Preparation and Feature Engineering

Data preparation followed the methodology outlined in the project requirements, beginning with cleaning and normalization of the client dataset. Missing values and duplicates were checked, date fields were converted into proper datetime format, and the age variable was derived from the cleaned birth-date field. The client data was then enriched using aggregated property-level features derived from sold and client-linked transaction records. This approach preserved a one-row-per-client structure while incorporating transaction-level intelligence.

Categorical variables such as client type, gender, country, region, acquisition purpose, loan application status, and referral channel were encoded into machine-readable form using one-hot encoding. Numerical variables including satisfaction score, age, sold property count, average sale price, and average floor area were scaled using standardization to ensure that large-value variables such as sale price did not dominate the clustering process. This preprocessing stage created a model-ready dataset suitable for unsupervised learning.

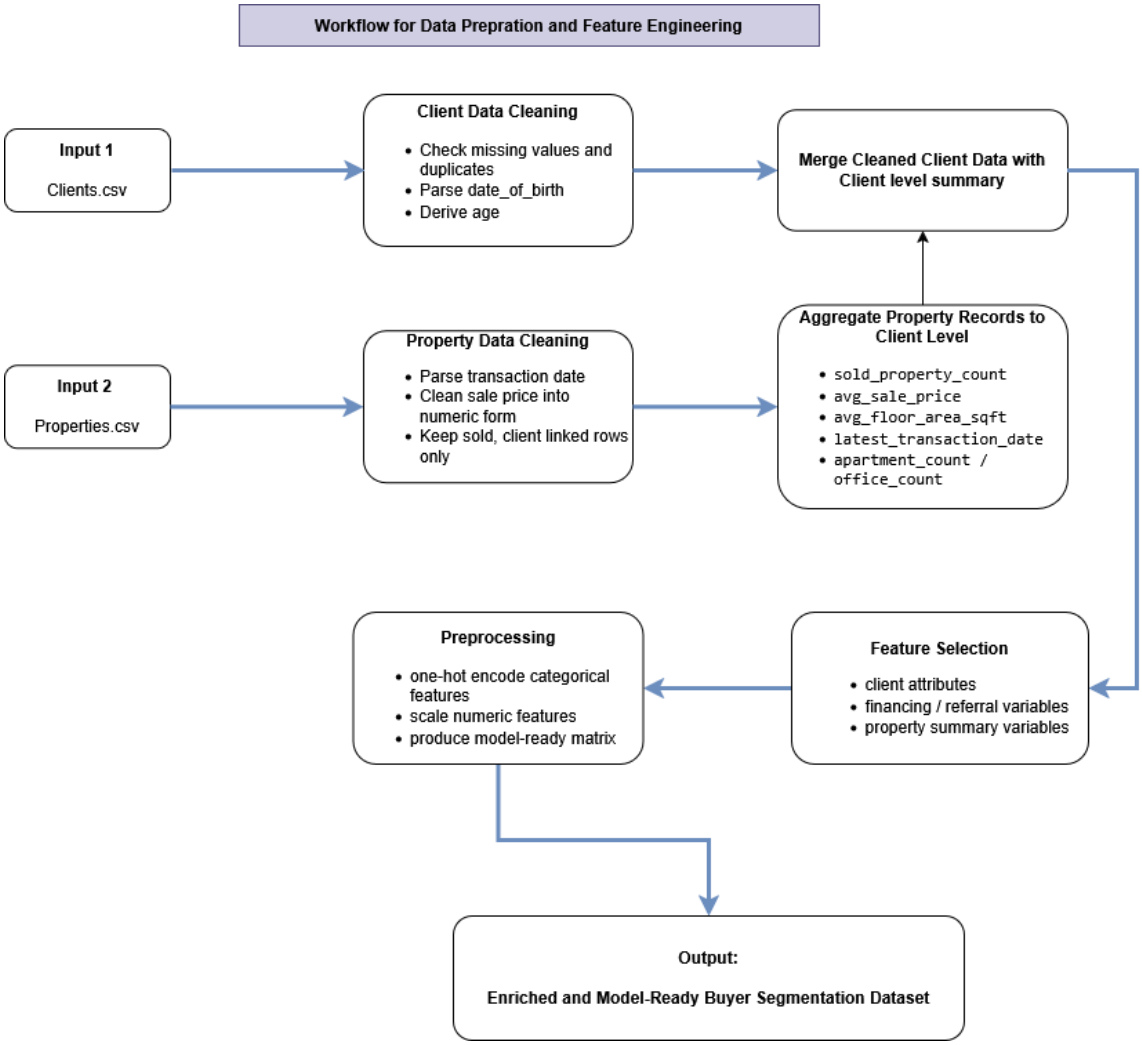

Figure 1 summarizes the data preparation and feature engineering pipeline used to transform raw client and property data into a model-ready dataset for clustering.

This workflow summarizes the preparation pipeline used in the study. Client-level data was cleaned and enriched using aggregated property-level transaction features, followed by feature selection, categorical encoding, and numeric scaling to create a model-ready dataset for clustering analysis.

5. Exploratory Data Analysis

The exploratory data analysis showed that the buyer base is dominated by individual clients rather than company clients. Home purchases were more common than investment purchases, and most clients had not applied for a loan. In addition, the website emerged as the largest referral source, followed by agencies, while direct client referrals were much smaller. These patterns suggest that the buyer base is primarily composed of mainstream residential clients rather than strongly differentiated investor or corporate categories.

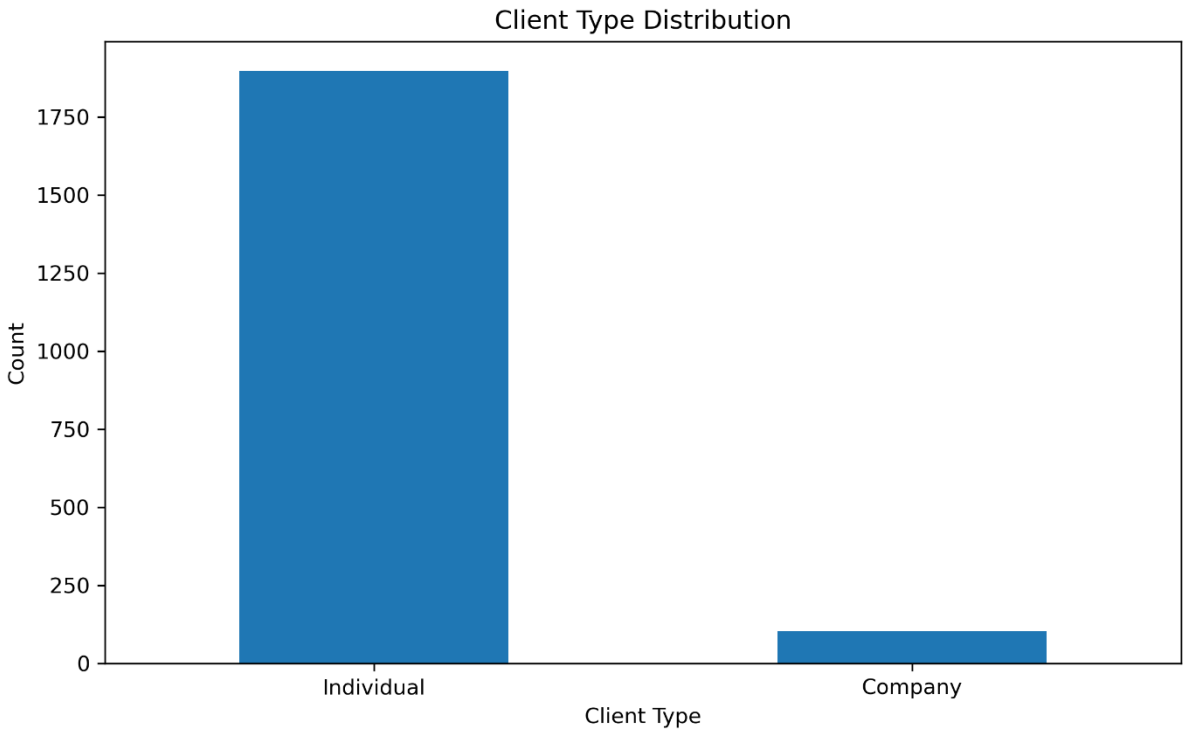

Figure 2 shows the strong dominance of individual clients in the dataset.

The distribution indicates that the platform primarily serves individual buyers, while company clients represent only a small minority of the overall buyer population.

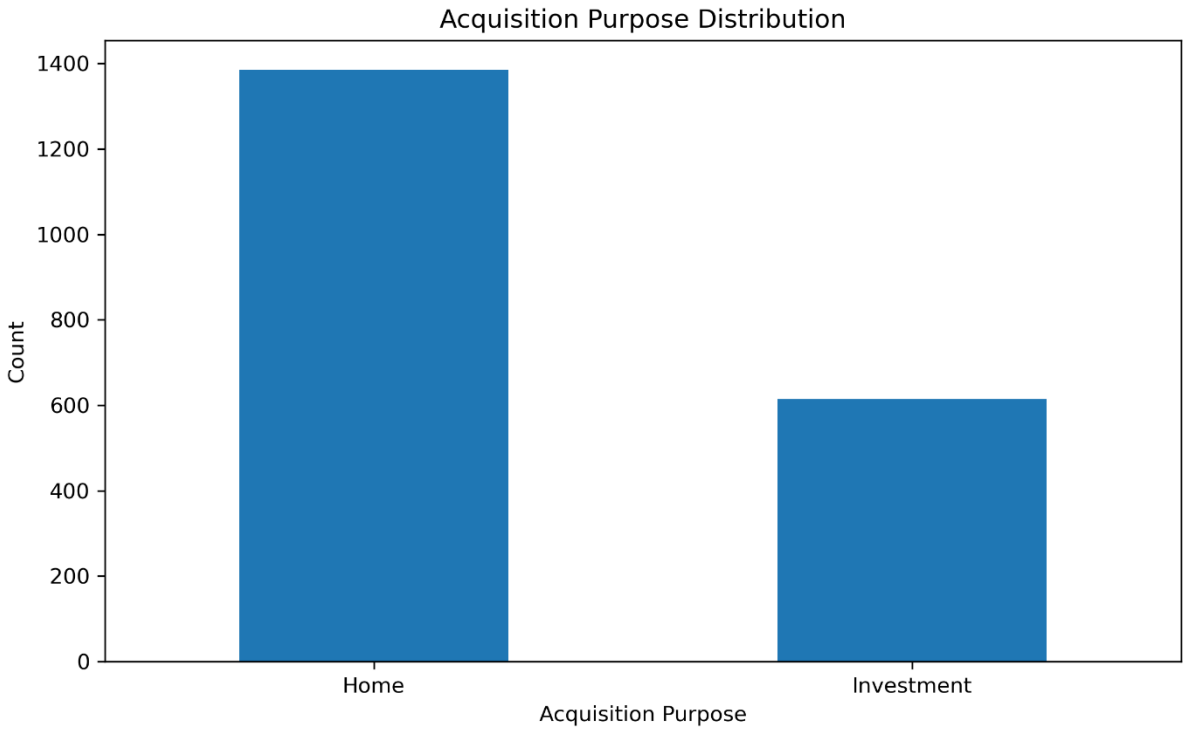

Figure 3 highlights the relative balance between home-oriented and investment-oriented purchases.

Home purchases dominate the dataset, although the investment segment remains large enough to justify targeted profiling.

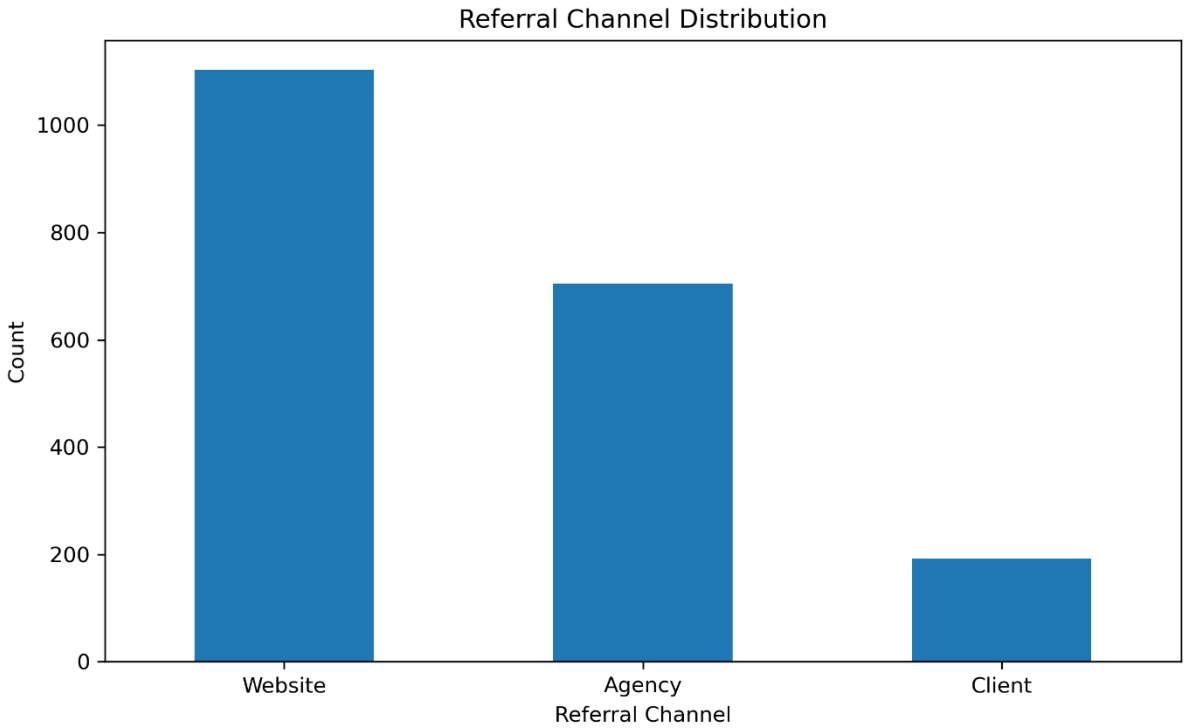

Figure 4 presents the main channels through which clients were acquired.

The website is the leading acquisition channel, with agencies also contributing a meaningful share, while direct client referrals remain comparatively limited.

Further analysis showed that some intuitive segmentation assumptions were weaker than expected. Loan applicants and non-loan applicants displayed a very similar split between home and investment purposes, suggesting that financing behavior alone does not strongly differentiate buyer intent in this dataset. Likewise, company clients showed only a modestly higher tendency toward investment purchases compared with individuals. These findings imply that acquisition purpose, client type, and financing status do not by themselves create sharply distinct segments.

Age-related comparisons were more revealing. While home buyers and investment buyers had very similar age distributions, individual clients were meaningfully older than company clients. This suggests that age may play a more useful role when combined with client type than when used to distinguish acquisition purpose on its own.

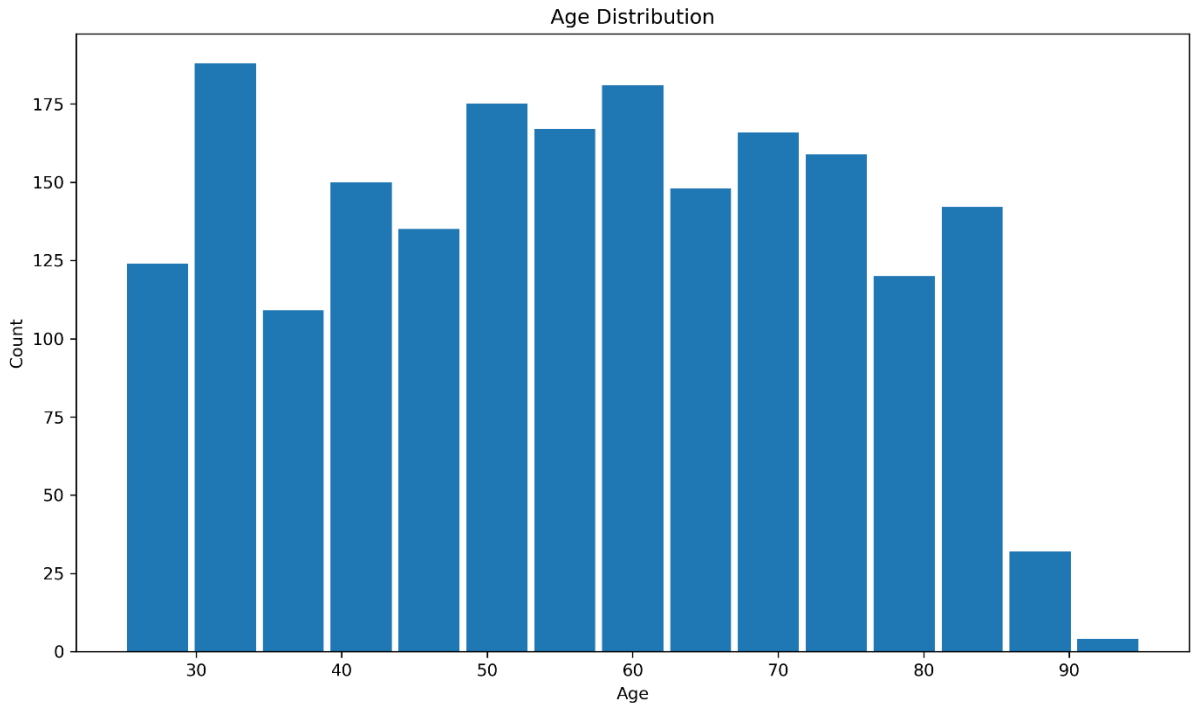

Figure 5 shows the overall age distribution of the buyer base.

The distribution is concentrated in middle-aged and older client groups, indicating that the dataset is not dominated by younger buyers.

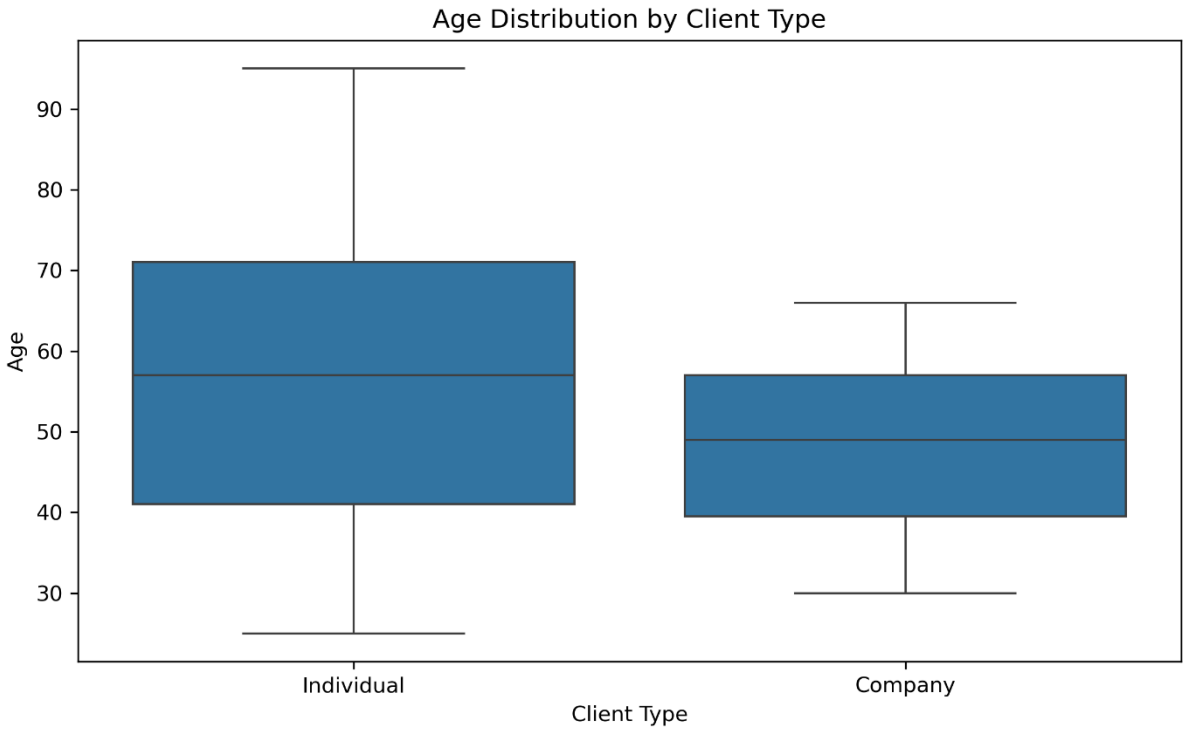

Figure 6 compares age distribution across client types.

The boxplot confirms that individual clients tend to be older and more widely distributed across age bands, whereas company clients are younger on average and more tightly concentrated.

At a broader level, the exploratory analysis suggested that the strongest structure in the data was likely to come not from basic descriptors such as client type or financing status alone, but from the interaction of demographics with property-level variables such as sale price, floor area, and number of sold properties. This justified the decision to enrich the client dataset using aggregated property features before clustering.

6. Clustering Methodology

The project followed the clustering workflow defined in the requirements document, which specifies data cleaning, feature encoding, feature scaling, K-Means clustering, hierarchical clustering, and evaluation using the Elbow Method and silhouette score. After cleaning and enriching the client dataset with aggregated property-level variables, the final modeling dataset included a mix of categorical and numerical features. Categorical variables such as client type, gender, country, region, acquisition purpose, loan status, and referral channel were transformed using one-hot encoding, while numerical variables including age, satisfaction score, sold property count, average sale price, and average floor area were standardized to place them on comparable scale.

K-Means clustering was selected as the primary segmentation method because it is computationally efficient and relatively easy to interpret, which makes it suitable for business-facing segmentation work. Hierarchical clustering was also applied as a validation method, in line with the project methodology, to examine whether a similar four-cluster structure could be observed under an alternative clustering approach.

To determine the number of clusters, multiple K values were tested and evaluated using both the Elbow Method and silhouette score. The elbow plot showed that model improvement continued as K increased, but the rate of improvement started to flatten around four to five clusters. The silhouette analysis suggested that a two-cluster solution offered the strongest mathematical separation, but four clusters emerged as the strongest non-trivial alternative and offered better business interpretability for segmentation purposes.

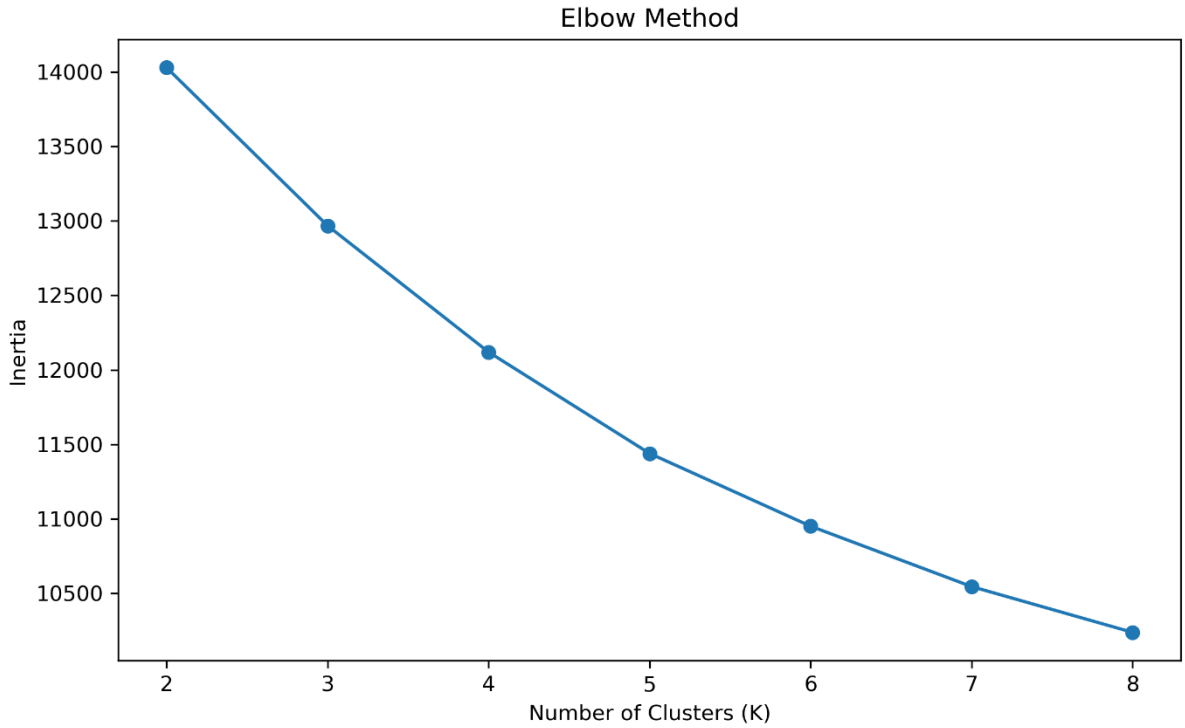

Figure 8 shows the decline in inertia as the number of clusters increases.

Although inertia improves steadily as more clusters are added, the curve begins to flatten around four to five clusters, suggesting diminishing returns beyond that range.

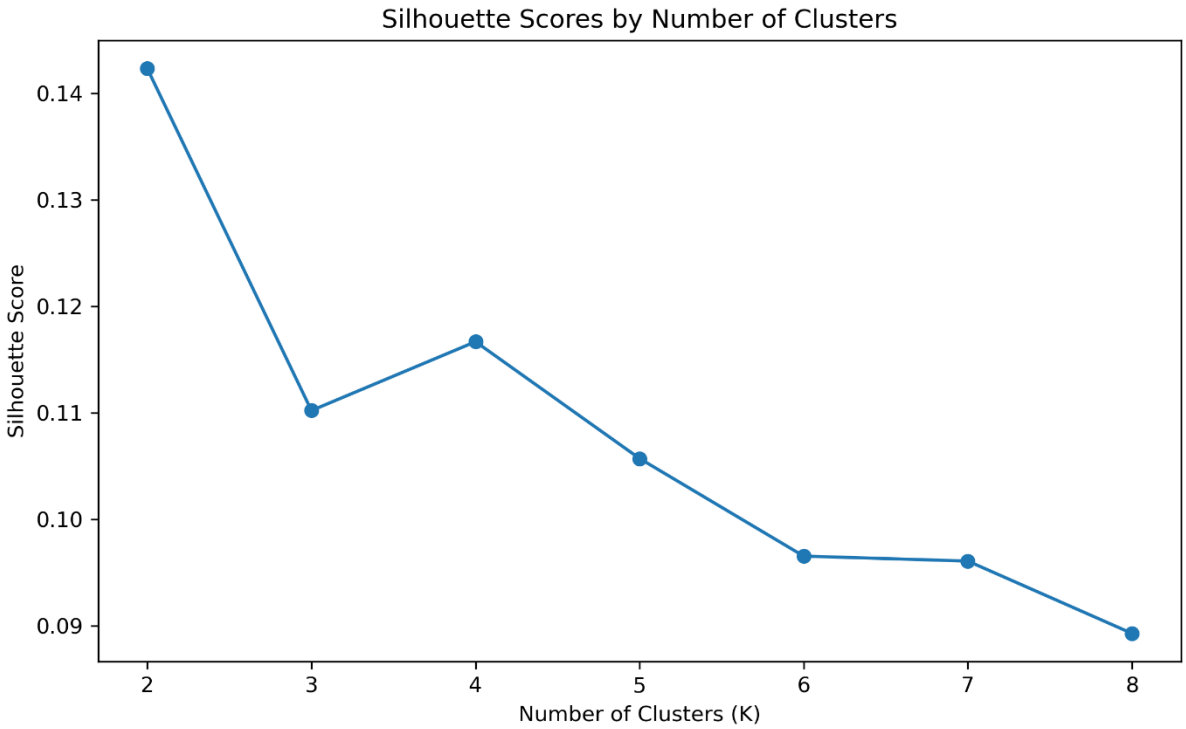

Figure 9 compares clustering quality across alternative values of K using silhouette score.

The highest silhouette score occurs at K = 2, but K = 4 provides the strongest balance between statistical performance and business usefulness among the richer segmentation options.

Based on this trade-off, K = 4 was selected as the primary clustering solution. This choice allowed the analysis to move beyond an overly broad two-group split while still remaining within the plausible range suggested by the elbow plot. The resulting K-Means clusters were then interpreted using both buyer attributes and property-related characteristics.

Hierarchical clustering was used to validate this decision. A sample-based dendrogram suggested a plausible four-branch structure, including one smaller niche branch alongside broader groups. When hierarchical clustering was run with four clusters on the full preprocessed dataset, the resulting size distribution was broadly similar to the K-Means solution, although the hierarchical silhouette score was lower. This indicated that K-Means provided the cleaner final segmentation model, while hierarchical clustering still supported the overall four-cluster interpretation.

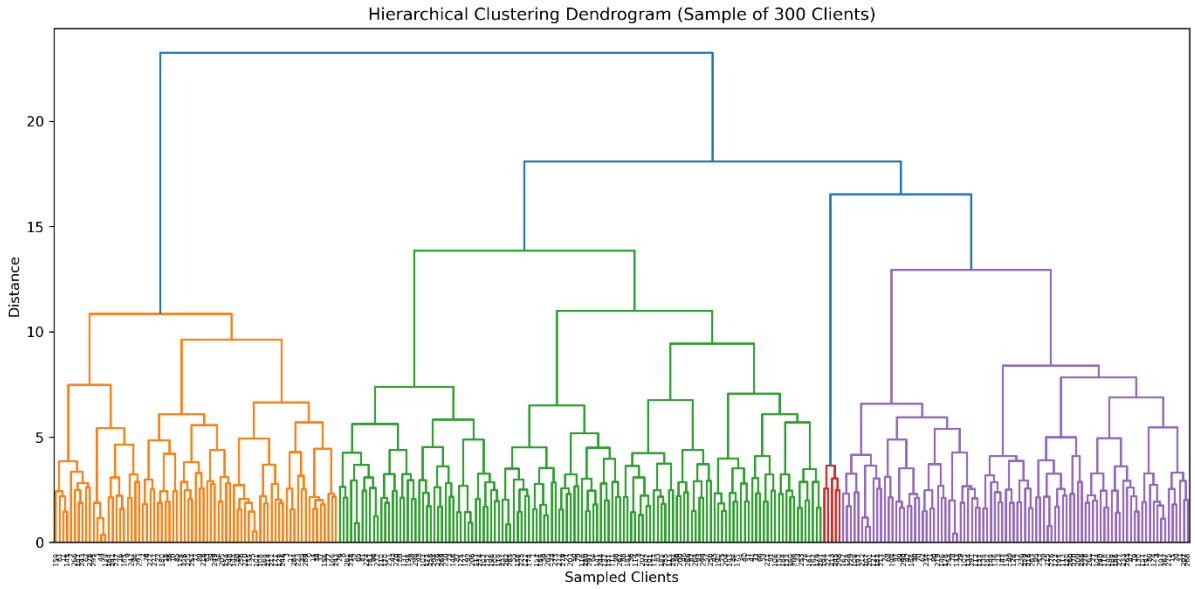

Figure 10 presents the hierarchical clustering dendrogram generated from a sample of 300 clients.

The dendrogram suggests a visually plausible four-cluster structure and supports the retention of K = 4 as the final segmentation choice.

7. Cluster Evaluation and Validation

The final K-Means model produced four clusters of unequal but usable size. Three clusters represented medium-to-large buyer groups, while one small niche cluster accounted for just over four percent of clients. The presence of a smaller specialized cluster was not, by itself, a weakness, especially because hierarchical clustering also identified a comparable niche segment. This strengthened confidence that the small cluster was not merely a random artifact of K-Means initialization.

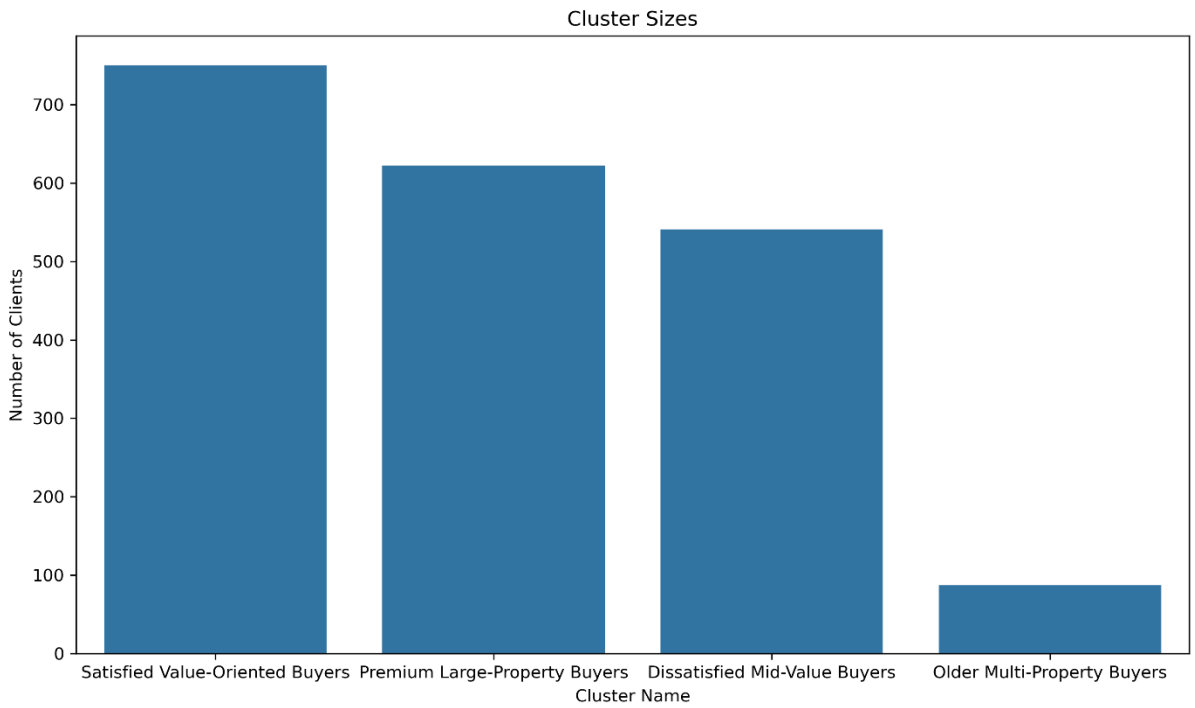

Figure 11 shows the relative size of the four identified buyer segments.

The segmentation contains three mainstream groups and one smaller niche cluster, which is consistent with both K-Means and hierarchical clustering outcomes.

The validation exercise also showed that the buyer groups are only moderately separated rather than sharply distinct. The K-Means solution outperformed the hierarchical solution on silhouette score for the four-cluster case, but the two methods still displayed broad overlap in group structure. This means the segmentation should be interpreted as a practical business framework rather than as a perfectly separated statistical taxonomy.

8. Cluster Profiles and Insights

The final K-Means model produced four practical buyer segments. While the PRD provides example segment types such as Global Investors, First-Time Buyers, Corporate Buyers, and Luxury Investors, the empirical clusters identified in this study only partially align with those archetypes. In this dataset, the strongest separation emerged from property value, floor area, transaction intensity, age, and satisfaction score rather than from client type, acquisition purpose, or loan usage alone. As a result, the final cluster names were based on observed behavior in the data rather than on forcing the sample into predefined labels.

Cluster	cluster_name	client_count	avg_age	avg_satisfaction	avg_sold_property_count	avg_sale_price	avg_floor_area_sqft	Cluster %
0	Premium Large-Property Buyers	622	55.26	3.08	3.42	427988.47	1402.26	31.1
1	Older Multi-Property Buyers	87	66.43	3.62	6.34	334367.82	1111.14	4.35
2	Dissatisfied Mid-Value Buyers	541	54.94	1.45	3.57	311577.8	1034.06	27.05
3	Satisfied Value-Oriented Buyers	750	55.78	4.06	3.59	307089.99	1022.2	37.5

Table 1. Summary profile of final buyer clusters

The table shows that the clusters differ most clearly in terms of average sale price, floor area, number of sold properties, age, and satisfaction, while other variables such as client type and financing behavior remain relatively similar across groups.

Cluster 0: Premium Large-Property Buyers

This segment represents approximately 31% of the buyer base and is characterized by the highest average sale price and the largest average floor area among the four groups. Buyers in this segment are of broadly average age relative to the full sample and do not stand out strongly on satisfaction or transaction volume. Their defining characteristic is the tendency to transact in higher-value, larger-unit properties. This segment can therefore be interpreted as a premium property-oriented buyer group rather than as a strongly distinct demographic or financing segment.

From a business perspective, this cluster is likely to be commercially important because it captures clients associated with larger transactions and higher-value property exposure. These buyers may respond well

to premium inventory recommendations, personalized advisory support, and campaigns focused on high-value or large-floor-area listings.

Cluster 1: Older Multi-Property Buyers

This is the smallest segment, accounting for just over 4% of clients, but it is also one of the most distinctive. Buyers in this cluster are substantially older on average and have the highest number of sold properties per client. Their average sale price is not the highest, but their transaction intensity clearly exceeds the rest of the sample. Satisfaction is also relatively strong compared with most other clusters.

This segment appears to represent an older, high-activity buyer niche, potentially corresponding to repeat purchasers, portfolio-style buyers, or more experienced real estate participants. Although the PRD mentions corporate and investor-style segments as desirable outputs, this group does not appear to be defined primarily by company status. Instead, its strongest markers are age and repeated transaction activity.

For Parcl, this is a strategically important niche. Even though it is small, it may justify targeted retention efforts, investor-style relationship management, or repeat-buyer engagement programs.

Cluster 2: Dissatisfied Mid-Value Buyers

This segment accounts for roughly 27% of the dataset and is defined most clearly by low satisfaction. In terms of age, transaction count, sale price, and floor area, it sits in the mid-range of the portfolio, but its satisfaction score is significantly lower than that of the other clusters. This makes it the clearest customer-experience risk segment in the model.

The cluster does not appear sharply different in terms of acquisition purpose, financing status, or client type, which means its value lies less in who the clients are and more in how they seem to be experiencing the buyer journey. This cluster is likely to be the most important from a service-improvement perspective.

For business strategy, this group should be treated as a service recovery priority. It may indicate friction in onboarding, matching quality, customer communication, or post-transaction support.

Cluster 3: Satisfied Value-Oriented Buyers

This is the largest segment, representing approximately 38% of the buyer base. It combines the highest satisfaction levels with lower average sale prices and smaller average floor area compared with the premium segment. Transaction counts are moderate, and age is broadly in line with the rest of the dataset.

This group appears to represent a mainstream value-oriented customer base that is relatively healthy from a customer-experience perspective. These buyers are not transacting at the highest value levels, but they are comparatively satisfied and therefore likely to be a stable operational segment.

For Parcl, this cluster may represent the best opportunity for scalable mainstream engagement. Efficient digital recommendations, continued service consistency, and retention-focused communication could all be relevant here.

Overall insight from the segmentation

Taken together, the four segments suggest that the buyer base is differentiated more strongly by **property economics and buyer experience** than by simple buyer-category labels. In other words, the clustering reveals more meaningful separation in:

- transaction intensity

- property value
- unit size
- age
- satisfaction

than in:

- client type
- acquisition purpose
- loan application behavior

This is an important result because it suggests that buyer segmentation in this market cannot be reduced to a simple split between home buyers and investors, or between individuals and companies. Instead, the evidence points toward a more nuanced structure driven by how clients participate in the market and what kinds of properties they transact in.

9. Business Recommendations

The PRD frames the project as a tool for improving marketing effectiveness, buyer targeting, investment insight, and property recommendation quality. The cluster results support those goals, but they suggest that the most useful interventions should be tailored by property behavior and customer experience rather than by broad demographic assumptions alone.

Recommendation 1: Prioritize premium targeting for Cluster 0

The Premium Large-Property Buyers segment should be used to support premium inventory positioning and high-value recommendation strategies. Since these clients are associated with the highest average sale prices and floor areas, Parcl should prioritize:

- premium listing recommendations
- larger-unit promotion
- tailored communication for high-value opportunities
- stronger sales support for large-ticket transactions

Recommendation 2: Build retention and loyalty strategies for Cluster 1

The Older Multi-Property Buyers segment should be treated as a strategic niche. Despite its small size, it is associated with the highest transaction intensity and therefore may represent disproportionate long-term value. Recommended actions include:

- investor-style retention programs
- repeat-buyer outreach
- account-based support or relationship management
- personalized alerts for multi-property opportunities

Recommendation 3: Treat Cluster 2 as a service-improvement priority

The Dissatisfied Mid-Value Buyers segment presents the clearest operational concern. Its poor satisfaction profile suggests that the customer experience for this group is weaker than for the rest of the buyer base. Parcl should investigate:

- support responsiveness
- recommendation relevance
- purchase journey friction
- communication gaps
- onboarding or follow-up quality

This cluster offers the strongest opportunity for experience-driven performance improvement.

Recommendation 4: Maintain and scale engagement for Cluster 3

The Satisfied Value-Oriented Buyers segment appears to be a stable, large, mainstream customer base. Since it combines lower-value average properties with relatively strong satisfaction, it may be well suited to:

- scalable digital engagement
- efficient recommendation engines
- mainstream retention campaigns
- cost-effective customer support models

Recommendation 5: Do not over-segment by client type or loan status alone

A major finding from this project is that client type, financing behavior, and acquisition purpose do not by themselves strongly differentiate the final clusters. This means Parcl should be cautious about building strategies that rely too heavily on simple labels such as “company buyer,” “loan applicant,” or “investment buyer” without considering property behavior and customer experience context.

10. Limitations

The clustering results should be interpreted with appropriate caution. First, the silhouette scores for the tested clustering models were moderate to weak, which suggests that the buyer groups are practical and usable rather than sharply separated. The model therefore provides a helpful segmentation framework, but not a perfectly clean taxonomy of buyer types.

Second, the final clusters were influenced more strongly by property-level features and satisfaction than by some of the client-level attributes highlighted in the original project framing. This means that although the PRD refers to segment types such as Global Investors, First-Time Buyers, Corporate Buyers, and Luxury Investors, the empirical data in this study only partially supported those archetypes.

Third, the dataset used here captures a limited snapshot of buyer behavior. Additional variables such as income, wealth level, financing amount, time-to-purchase, property location precision, and post-sale engagement could potentially improve segmentation quality in future work.

Finally, region and country variables introduced a relatively high number of encoded features, which may have added some noise into the clustering structure. A future refinement could test simplified geography variables or alternative clustering feature sets.

11. Conclusion

This project developed a machine learning-based buyer segmentation framework for real estate market intelligence using client-level data enriched with aggregated property transaction features. The final analysis used K-Means clustering as the primary segmentation model, supported by hierarchical clustering as a validation method, in line with the methodology described in the project requirements.

The results indicate that buyer segmentation in this dataset is driven more strongly by property value, floor area, transaction intensity, age, and satisfaction than by basic buyer labels such as client type, purchase purpose, or financing behavior alone. Four practical segments were identified: Premium Large-Property Buyers, Older Multi-Property Buyers, Dissatisfied Mid-Value Buyers, and Satisfied Value-Oriented Buyers. Although the statistical separation between groups was moderate rather than sharp, the clusters still provide a useful framework for improving marketing, recommendation strategy, retention, and customer experience management.

Overall, the project demonstrates how clustering-based buyer intelligence can support more targeted and data-driven real estate decision-making. For Parcl, the main value lies not in discovering perfectly pure buyer types, but in building a practical segmentation system that supports smarter commercial action.